

Market Structure Analysis Framework (Progress + Next Steps)

We are building a market structure analysis framework based on rolling-window statistical behavior of price data. The goal is to move from raw indicators to **regime-based structural interpretation of markets**.

✓ Current Progress

We have defined and implemented the following core metrics:

1. Scale (Movement Energy)

- Rolling mean of absolute price changes:

$$scale = E(|close_t - close_{t-1}|)$$

- Represents **typical movement energy / activity level**
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2. Central Tendency Distortion (C)

$$C = \frac{|mean - median|}{scale}$$

- Measures **structural deformation of the distribution**
 - Interprets how much the distribution center is skewed relative to movement energy
 - High C → structural instability or regime shift
 - Low C → stable equilibrium structure
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3. Symmetry (S)

$$S = \frac{|n_{above} - n_{below}|}{n_{above} + n_{below}}$$

- Measures **directional imbalance inside a rolling window**
 - Low S → balanced behavior (equilibrium-like)
 - High S → directional dominance
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4. Behavioral Extensions

- Imbalance

$$|n_{above} - n_{below}|/w$$

→ local directional bias strength

- **Reversion**

$$|close - mean|/scale$$

→ strength of mean reversion pressure

Key Conceptual Shift

We are no longer treating metrics as independent indicators.

Instead:

We are building a **market state system** composed of structural, energetic, and behavioral dimensions.

Current Visualization Layers

1. **Time Series Layer**

- C (structural drift)
- distortion

2. **Structure Space**

- scale vs distortion relationship

3. **Behavior Layer**

- imbalance
- reversion

Open Conceptual Question

We began exploring extending Symmetry (S) into a contextual regime variable:

Proposed extension:

$$S_z = \frac{S - mean(S)}{std(S)}$$

This would allow classification into regimes:

- Balanced (equilibrium)

- Rising pressure
- Dominance / breakout

However, this is currently **not implemented and conceptually paused**.

Current Status

We are currently paused at:

deciding whether to extend S into a regime classifier using normalization (S_z), and whether to build a full state machine combining S, C, scale, imbalance, and reversion.